ACCELERATOR PEDAL

REMOVAL

The accelerator pedal is connected to the throttle body linkage by the throttle cable. The cable is protected by a plastic sheathing and is connected to the throttle body linkage by a ball socket. It is connected to the upper part of the accelerator pedal arm by a plastic retainer (clip) (Fig. 1). This retainer (clip) snaps into the top of the accelerator pedal arm. Retainer tabs (built into the cable sheathing) (Fig. 1) fasten the cable to the dash panel.

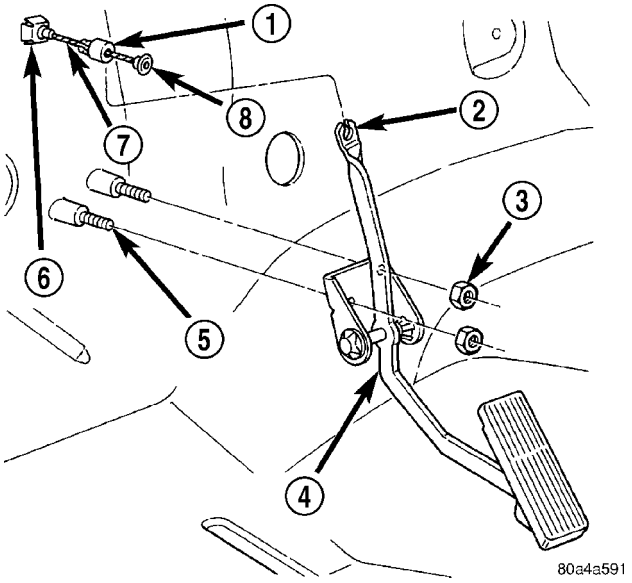

Fig. 1 Accelerator Pedal Mounting

1 - CABLE RETAINER
2 - SLOT
3 - MOUNTING STUDS (2)
4 - PEDAL/BRACKET ASSEMBLY
5 - MOUNTING STUDS
6 - RETAINER TABS
7 - CABLE
8 - CABLE STOP

CAUTION: Be careful not to damage or kink the cable core wire (within the cable sheathing) while servicing the accelerator pedal or throttle cable.

(1) From inside vehicle, hold up accelerator pedal. Remove plastic cable retainer (clip) and throttle cable core wire from upper end of accelerator pedal arm (Fig. 1). Plastic cable retainer (clip) snaps into pedal arm.

(2) Remove accelerator pedal mounting bracket nuts.

(3) Remove accelerator pedal assembly.

INSTALLATION

(1) Place accelerator pedal assembly over mounting studs protruding from floor pan. Tighten mounting nuts to 8.5 N·m (75 in. lbs.) torque.

(2) Slide throttle cable into opening (slot) in top of pedal arm. An index tab is located on pedal arm. Rotate and push plastic cable retainer (clip) into accelerator pedal arm opening until it snaps into place on index tab.

(3) Before starting engine, operate accelerator pedal to check for any binding.

CRANKSHAFT POSITION SENSOR

DESCRIPTION

2.4L

The Crankshaft Position (CKP) sensor is mounted into the right front side of the cylinder block. It is positioned and bolted into a machined hole.

4.0L

The Crankshaft Position (CKP) sensor is located near the outer edge of the flywheel (starter ringear).

OPERATION

2.4L

Engine speed and crankshaft position are provided through the CKP (Crankshaft Position) sensor. The sensor generates pulses that are the input sent to the Powertrain Control Module (PCM). The PCM interprets the sensor input to determine the crankshaft position. The PCM then uses this position, along with other inputs, to determine injector sequence and ignition timing.

The sensor is a hall effect device combined with an internal magnet. It is also sensitive to steel within a certain distance from it.

A tonewheel (targetwheel) is a part of the engine crankshaft (Fig. 2). This tonewheel has sets of notches at its outer edge.

The notches cause a pulse to be generated when they pass under the sensor. The pulses are the input to the PCM.